

Internship Report

Task 1

Task: Responsive Coffee Shop Landing Page using HTML & CSS

Today, I developed a responsive Coffee Shop Landing Page using HTML5 and CSS3. The website includes a Header with navigation links, a Hero section with a welcoming message and call-to-action button, an About section describing the coffee shop, a Menu section displaying three coffee items, a Contact section with a simple form, and a Footer containing copyright information.

I used CSS Flexbox and Grid to create a clean and organized layout. To improve the user experience, I added hover effects to buttons, navigation links, and menu cards. I also implemented responsive design using media queries so that the website adapts properly to desktop, tablet, and mobile screen sizes.

During this task, I improved my understanding of HTML page structure, CSS styling, responsive layouts, and modern web design practices. I tested the website on different screen sizes to ensure a consistent user experience. The completed project has been uploaded to GitHub with proper version control.

Task 2

Task: Add JavaScript: navbar, form validation, slider, dark mode, scroll animations.

Today, I enhanced my Coffee Shop Landing Page by adding multiple interactive features using JavaScript, making the website more dynamic, user-friendly, and visually appealing. I successfully implemented a sticky navigation bar that remains fixed at the top of the page while scrolling, allowing users to access different sections of the website easily.

I also added form validation to the Contact section. The form now checks whether the Name, Email, and Message fields are filled correctly before submission. If any field is empty or the email format is invalid, an appropriate error message is displayed. After successful validation, a confirmation message is shown and the form is reset automatically.

To improve the visual experience, I developed a Hero Image Slider with Previous and Next navigation buttons. The slider also changes images automatically every three seconds, making the homepage more attractive and interactive.

Another feature I implemented is Dark Mode, which allows users to switch between light and dark themes with a single click. This changes the background, text, navigation bar, cards, and footer, providing a better viewing experience in different lighting conditions.

Finally, I integrated the AOS (Animate On Scroll) library to add smooth scroll animations to the Hero, About, Menu, Contact, and Footer sections. These animations improve the overall user experience and give the website a modern appearance.

Throughout the development process, I tested each feature individually to ensure proper functionality and responsiveness across different screen sizes. After completing all the enhancements, I verified that the website worked correctly without affecting the existing HTML and CSS design.

Technologies Used: HTML5, CSS3, JavaScript, AOS (Animate On Scroll) Library, Git, GitHub, and Visual Studio Code.

Task 3

Task: Build frontend in React with reusable components and React Router.

Work Completed

- Created a new React project using Create React App.
- Organized the project into reusable components:
 - Navbar
 - Hero
 - About
 - Menu
 - Contact
 - Footer
- Created a Home page by combining all the components.
- Implemented React Router for page routing.
- Reused the existing CSS and images in the React project.
- Developed the Hero Image Slider using React Hooks (useState and useEffect).
- Implemented Dark Mode using React state.
- Added Contact Form Validation in React.
- Integrated AOS (Animate On Scroll) library for scroll animations.
- Fixed layout and styling issues to maintain responsiveness.
- Updated the project folder structure and README documentation.
- Prepared the React project for GitHub upload.

Technologies Used

- React.js
- JSX
- CSS3
- JavaScript (ES6)
- React Router DOM
- React Hooks (useState, useEffect)
- AOS Library
- Git & GitHub

Outcome

Successfully converted the Coffee Shop Landing Page into a React application with reusable components, routing, image slider, dark mode, form validation, and scroll animations. The application is responsive and ready for the next phase of backend development.

Task 4

Task: Develop Node.js & Express REST APIs for registration, login and contact.

Work Completed

- Created a Backend project using Node.js.
- Installed Express.js, CORS, and Nodemon.
- Configured the Express server.
- Created folders for routes, controllers, and data.
- Developed REST APIs for:
 - User Registration
 - User Login
 - Contact Form
- Added basic validation for API requests.
- Tested all APIs successfully using Postman.
- Verified the backend server on **http://localhost:5000**.

Technologies Used

- Node.js
- Express.js
- JavaScript
- Postman

Outcome

Successfully developed and tested the Registration, Login, and Contact REST APIs using Node.js and Express.js. The backend server is working properly and is ready for MongoDB integration in the next phase.

Task 5

Task: Integrate MongoDB, CRUD operations, connect frontend using Axios/Fetch.

Tasks Completed

- Connected the backend to **MongoDB Atlas**.
- Implemented **User Registration API** and **Login API**.
- Created **Contact API** with full **CRUD operations** (Create, Read, Update, Delete).
- Tested all APIs successfully using **Postman**.
- Connected the **React Registration Page** to the backend using **Axios**.
- Connected the **React Login Page** to the backend using **Axios**.
- Connected the **React Contact Form** to the backend using **Axios**.
- Successfully stored **user details** and **contact messages** in **MongoDB Atlas**.
- Updated the **Navbar** with Login and Register options.
- Updated the project **README.md** with the latest completed features.

Technologies Used

- React.js
- Node.js
- Express.js
- MongoDB Atlas
- Mongoose
- Axios
- bcryptjs
- Postman

Outcome

Successfully completed MongoDB integration, CRUD operations, and frontend-backend communication. The Coffee Shop application now works as a full-stack MERN project with user authentication and contact message storage.